MYSURU ROYAL INSTITUTE OF TECHNOLOGY

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CSE

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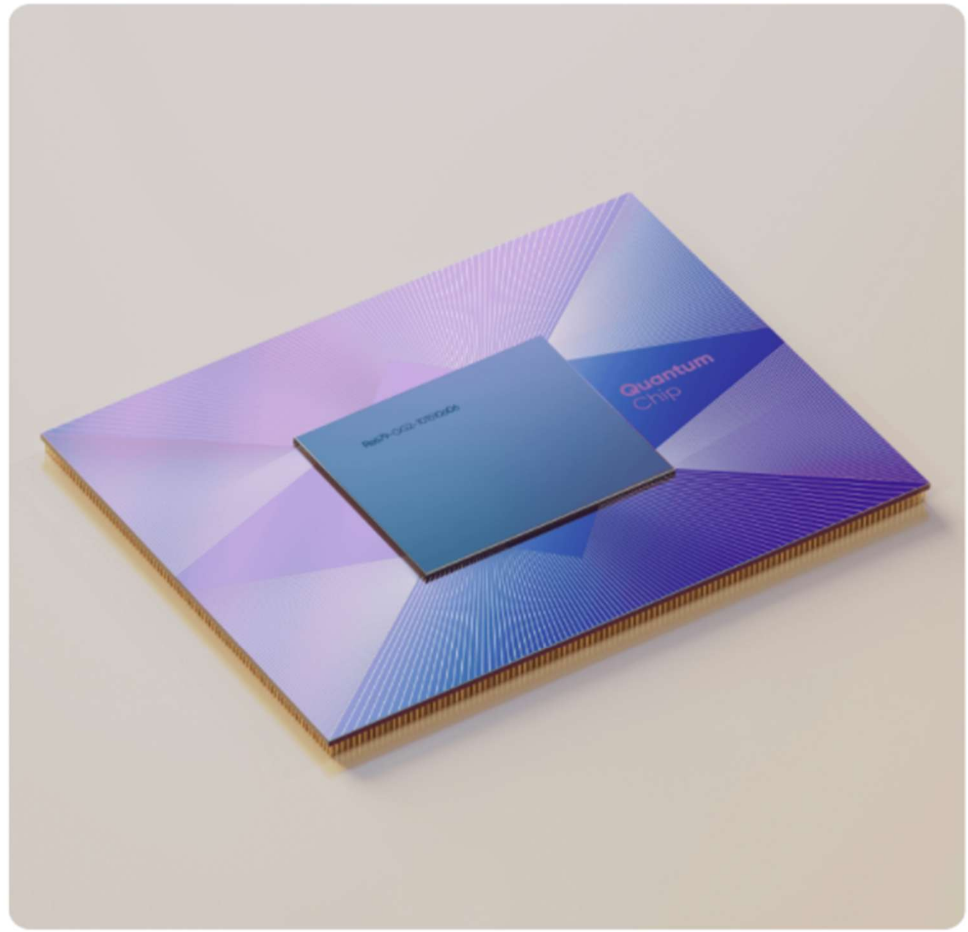
CSE

User Datagram Protocol



Fast, Connectionless Data Transmission Protocol

Contents

User Datagram

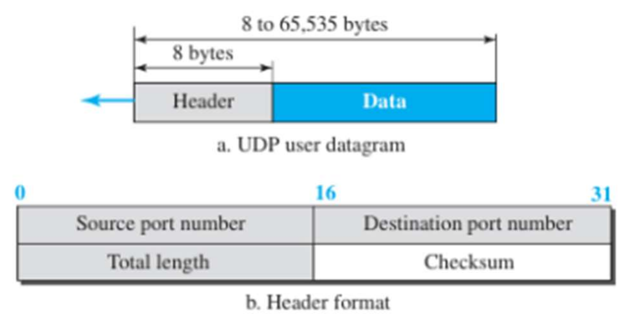
Definition and Characteristics of User Datagram

User Datagram is a connectionless, unreliable packet delivering data without guaranteed order or delivery confirmation.

Datagram Structure and Format

UDP datagram consists of a header with source, destination ports, length, and checksum fields, followed by data payload.

Figure 24.2 *User datagram packet format*



UDP Services Overview

Connectionless Service

Stateless Communication Model

UDP operates without establishing a dedicated end-to-end connection, enabling communication without prior handshaking.

Minimal Overhead and Fast Data Transfer

Lacking connection setup phases and acknowledgments reduces latency, making UDP suitable for time-sensitive applications.

Lack of Reliability and Flow Control

UDP does not guarantee packet delivery, order, or error correction, placing responsibility on the application layer.

Broadcast and Multicast Support

UDP supports transmission to multiple recipients through broadcasting and multicasting, facilitating efficient group communication.

Datagram Transmission and Structure

Data is sent in discrete packets called datagrams, each containing source and destination port information for delivery.

UDP in Peer-to-Peer Communication

Asynchronous Transmission:

UDP enables non-blocking communication by sending data without waiting for acknowledgments, which supports real-time applications and reduces latency in peer-to-peer networks.

Custom Reliability Mechanisms:

Since UDP does not guarantee delivery, peer-to-peer systems often implement their own error checking, retransmission strategies, or sequencing to ensure reliable data exchange over UDP.

Multicast & Broadcast Support:

UDP supports efficient data distribution to multiple peers simultaneously using multicast or broadcast, making it suitable for group communications in peer-to-peer environments.

NAT Traversal via Hole Punching:

UDP facilitates NAT traversal by allowing peers behind routers to create temporary mappings through hole punching, enabling direct communication without requiring centralized servers.

Flow Control

UDP does not implement any flow control mechanism, meaning it does not regulate the rate at which data is sent between sender and receiver. This can lead to potential data loss if the receiver cannot process incoming packets fast enough.

No Built-in Flow Control

Since UDP does not manage flow, the receiver's buffer can overflow when flooded with messages, resulting in packet drops. This makes it unsuitable for applications requiring guaranteed delivery or strict reliability.

Risk of Receiver Overflow

Because UDP lacks flow control, applications using UDP must implement their own methods to avoid overwhelming the receiver, such as by limiting sending rate or managing buffer capacity.

Responsibility on Application Layer

The absence of flow control simplifies UDP's design and reduces protocol overhead, making it ideal for latency-sensitive or multicast applications where speed is prioritized over guaranteed delivery.

Error Control

Error Control There is no error control mechanism in UDP except for the checksum.

This means that the sender does not know if a message has been lost or duplicated.

When the receiver detects an error through the checksum, the user datagram is silently discarded.

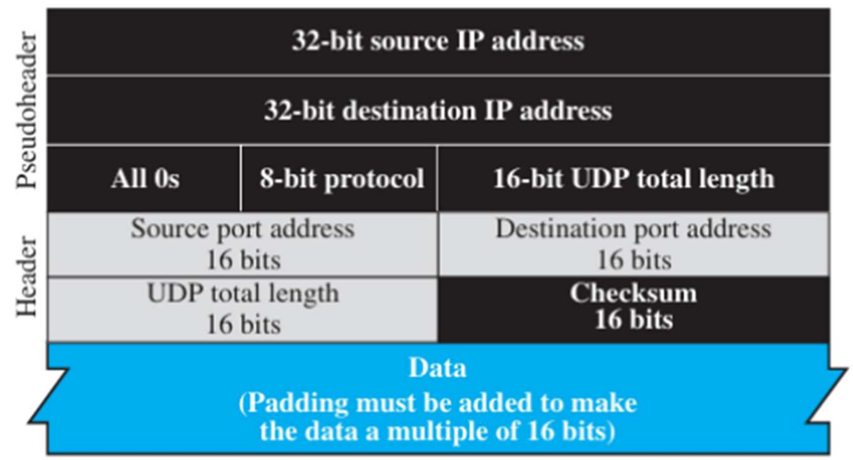
The lack of error control means that the process using UDP should provide for this service, if needed.

Checksum

Purpose of Checksum in UDP

Provides error detection to ensure data integrity during transmission. It helps identify corrupted segments in UDP datagrams.

Figure 24.3 *Pseudoheader for checksum calculation*



Applications of UDP

- Simple request-response communication → Example: DNS
- Internal flow/error control mechanisms → Example: TFTP
- Multicasting support → Embedded in UDP, not in TCP

- Management processes → Example: SNMP
- Route updating protocols → Example: RIP
- Interactive real-time applications → Example: Skype, streaming

Thank you